Arya Sapkal

929-206-6801 | arya.sapkal1@gmail.com | linkedin.com/in/aryasapkal | github.com/AryaSapkal

Education

Rutgers University–New Brunswick New Brunswick, NJ
B.S. in Computer Science *Expected May 2028*

University at Albany Albany, NY
Computer Science, GPA: 3.95 *Aug 2024 – May 2026*

- Honors College, Dean's List, Spellman Scholar
  Coursework: Algorithms, Data Structures, Operating Systems, Computer Architecture, System Fundamentals

Experience

Undergraduate Machine Learning Research Intern May 2026 – Present
UAlbany Air Quality Lab *Albany, NY*
About: Analyzing indoor/outdoor air quality factors in New York for decision-makers using machine learning.

- Clean and preprocess 60+ questionnaires with 300+ variables each for longitudinal air quality machine learning by constructing an end-to-end data workflow with Python and Pandas
- Process and analyze 10 million+ rows of particulate matter time-series data across 200+ sensors for new insights
- Save 20+ hours and automate questionnaire data collection by replacing paper forms with integrated digital forms
- Collaborate with 3+ team members through meetings, digital tools, and documentation

Undergraduate Researcher Mar 2026 – May 2026
UAlbany Data Management and Mining Lab *Albany, NY*

- Debugged and analyzed a 1,500+ line research codebase with 2,200+ DNA sequence samples for bioinformatics
- Refactored legacy code and expanded documentation to improve maintainability and team efficiency

Undergraduate Researcher Oct 2025 – Dec 2025
UAlbany Computer Vision and Machine Learning Lab *Albany, NY*

- Enhanced infant vital-sign detection pipelines (heart rate, sleep state) using 160+ 1-minute video recordings
- Used Python, OpenCV, MediaPipe, PyTorch, and Pandas for data processing and modeling

Software Application & Business Development Intern Jan 2025 – May 2025
ReWire Energy *Albany, NY*

- Designed a Figma onboarding prototype with 7+ screens and authored a 13-page front-end specification document
- Conducted broad competitive analysis across 150+ companies in AI and grid resiliency
- Presented project findings with 3 teammates to 60+ attendees at the UAlbany Showcase via NSF I-Corps Albany

Projects & Initiatives

Operating System Simulator | *Java* Feb 2026 – Present

- Engineered a multithreaded OS simulator with scheduling, virtual memory, messaging, and device management

CPU Simulator | *Java* Feb 2026 – Present

- Developed a CPU simulator with 20+ instructions and a processor, memory, assembler, and bit-level operations

Battleship (Networked Game) | *C, TCP Sockets* Sep 2025 – Dec 2025

- Built a two-player interactive Battleship game with TCP socket networking

Computer Vision Eye Contact Model | *PyTorch* Mar 2025 – Apr 2025

- Built an eye-contact detection model using 140+ labeled images

Hackathons

- PennApps XXVI (Top 3 of 27, Cerebras Track): Built a carbon-awareness app using Python with 3 team members
- Built AI, healthcare, and educational software projects at HackPrinceton, Columbia DevFest, and Cornell AI Hackathon using Python, Flask, FastAPI, TypeScript, Firebase, and Figma

Technical Skills

Languages: Java, Python, C, JavaScript
Libraries/Frameworks: PyTorch, Pandas, Matplotlib, SQLite, OpenCV, FastAPI, React
Tools: Git, GitHub, Linux/Unix, Azure DevOps, Google Colab, Figma
Other: HTML, CSS, Technical Writing, Spanish (basic), Taekwondo (black belt), Sci-Fi writing (200+ pages)